VIP PROJECT: CREATING ACCESSIBLE MATHEMATICS AND STATISTICS RESOURCES 2025 SEMESTER 1 PROJECT

STARMAST Syllabus Mapping Handover Document:

School of Mathematics and Statistics

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Martinmas Semester: December 2025

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Project brief

This handover document will breakdown down the problem, solution, and benefits of the of the newly developed navigation page to allow students and teachers globally to access materials that align to their course requirements in a simplified way.

Problem Statement

Currently, STARMAST resources can be accessed via the website's search engine, index or home page, making it challenging for users to identify materials that align with their course requirements and thus limiting ease of use and global adoption or the website.

Solution and Benefits

The project aims to produce a new interactive navigation system, allowing to find relevant STARMAST resources by based on what they are studying through user selection of country, exam board, course, and topic.

This new method of content management would allow for efficient access to relevant material as users will be able to quickly find guides that match their syllabus and clearer labelling will help students and teachers have clarity as to how the content STARMAST has fits within their learning pathway. Furthermore, this would allow for greater ease of global use with materials becoming more universally applicable and easier to adopt across diverse teaching contexts and school systems. There is also the potential for this to benefit the STARMAST team, with the standardised categorisation making it easier to see where topic gaps lie. This will assist in prioritising topics for guide and factsheet development.

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Scope of work currently completed

Dataset structure

The csv file to evolve the with headings {Country, Exam board, Course, Topic, Link}:

- Country: Country of syllabus
- Exam board: Awarding body (for example: AQA)
- Course: To limit curricula (for example: A-level Mathematics)
- Topic: List of topics covered by the course
- Link: To list STARMAST courses related to the topic using file name from "starmast-master" folder

File location

The current Excel and CSV file is under the name, "finalised_csv_content_management_final". Please speak to STARMAST Coordinator, Thomas Coleman (tdhc@st-andrews.ac.uk), to receive access to the file.

Webpage structure and logic

The following figure highlights the planned logic for this new navigation page:

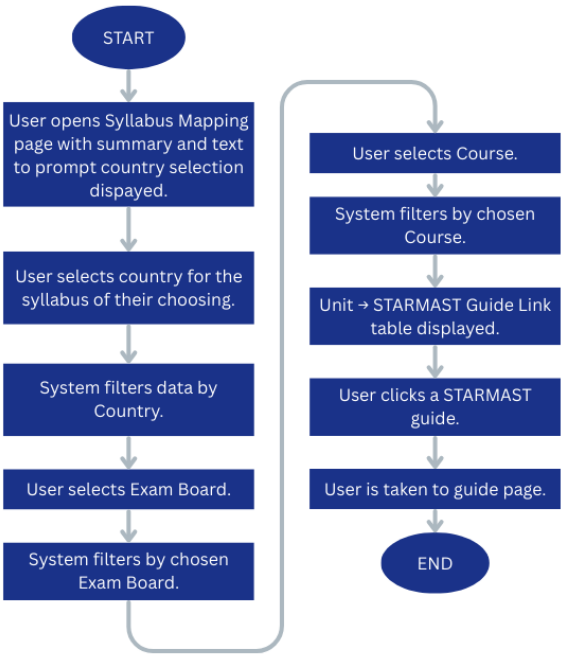

Figure 1: Flowchart highlighting logic for the syllabus mapping navigation page.

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Future directions and outstanding tasks

Recommendations of where to go with the current standing of the current project:

1. Add Syllabus Index to the navigation bar.
2. Extend to further countries, exam boards, and courses.
3. Extend to University of St Andrews courses beginning with "PH1011: Physics 1A" as already mapped by Ali Kikivarakis (see file "usta_physics_courses_mapping.xlsx").
4. Adding a message such as "No guides available for this section" for units with no guides mapped (see known issue 2).
5. Highlight new guides once filtering has occurred.
6. Allowing personalised navigation or saved pathways so users can save their country/exam board/course preferences without having to re-select when revisiting the site.
7. Enhance accessibility by allowing for full keyboard navigation.
8. Incorporate advancements made this system such as:
    a. Allow for multi-language support for the navigation system.
    b. Using analytics to track most selected countries, most accessed courses, and thus most visited guides from the system – valuable for prioritising future guide and content creation.

Known issues/limitations

Most limitations stem on the navigation system's dependence on the structure and consistency of the content management CSV file. See the following for specifics:

1. The mapping has been arranged using the file directory so if file names are changes the csv will need to be updated.
2. Missing guides for certain units means that units won't show on the directory (for example: Australia -> QCAA -> Specialist Mathematics -> Unit 4) does not show as there are no relevant guides for this unit).
3. As STARMAST is upscaled and the CSV becomes bigger the performance of the JavaScipt may degrade, therefore optimisation may be required in the future.
4. There are no error tolerances so small inconsistencies in the CSV will break the code.

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Contact details

Please reach out to Anja Colombi (ac457@st-andrews.ac.uk) if there are any questions regarding this project or information in this document.

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